



澳門理工大學
Universidade Politécnica de Macau
Macao Polytechnic University

應用科學學院
Faculdade de Ciências Aplicadas
Faculty of Applied Sciences

Doctor of Philosophy in Computer Applied Technology

Thesis Defence Examination

Thesis Review Comments

Student

Name	Na Han	ID	P2314691
Supervisor	Dr. Yapeng Wang		
Title of Thesis			
Date of Submission	2026-06-04		

Reviewer

Name	Prof. Lin Chen	Date	2026-06-12
------	----------------	------	------------

Recommendation

☒ Accepted ☐ Suggestion for Revision*

* "Suggestion for Revision" will result in the delay of the oral defence.

Comments

Overview.

This thesis focuses on developing deep learning-based intelligent diagnostic solutions for chest CT images, targeting early screening and risk prediction of thyroid cancer and pulmonary nodules, two high-burden clinical diseases. The author designs two core models: the DVT model for thyroid cancer screening based on Transformer and secondary transfer learning, and the CL-IJLO framework for longitudinal pulmonary nodule malignancy prediction combining contrastive learning and improved joint loss optimization. An exploratory multimodal fusion study on lung tumor subtype classification is also included. All models are trained and verified on multi-center clinical CT datasets from different regions. Experimental results show competitive overall performance across multiple evaluation metrics. The research traces a technical route from model design to multi-group validation, aiming to address clinical pain points such as heavy physician workload and subjective diagnosis, and explores the application value of AI in routine chest CT auxiliary diagnosis.

Knowledge Base and Literature Review.

The literature review covers domestic and overseas research on medical image analysis, deep learning applications in thyroid and lung cancer diagnosis, transfer learning and sequence modeling technologies. The author surveys the evolution of traditional radiomics and mainstream deep network architectures, and summarizes the strengths and existing defects of existing diagnostic models. Typical research results and representative algorithms are cited adequately. Meanwhile, the review identifies key challenges including medical data scarcity, CT feature redundancy and mismatched model optimization goals with clinical demands. Overall, the literature review supports the subsequent research design.

Originality and Independence.

The research is completed independently by the author, with targeted improvements made on existing deep learning methods. For thyroid cancer screening, a two-stage secondary transfer learning strategy is proposed to adapt Transformer to small-sample, noisy chest CT data. For pulmonary nodule prediction, a contrastive learning-based feature compression module and improved joint loss function are designed to reduce data redundancy and optimize model training. The multimodal fusion part is an exploratory attempt for tumor subtype classification. The author independently completed dataset collection, model building, ablation and comparative experiments, reflecting basic independent research ability.

Arguments and Reasoning.

The thesis follows a standard logic of raising problems, proposing solutions and conducting experimental verification. Each model design corresponds to specific clinical and technical difficulties, and the author explains the design ideas of network structures, loss functions and training strategies in detail. Experimental conclusions are supported by quantitative indicators such as accuracy, AUC, sensitivity and specificity, as well as ablation tests to verify the effectiveness of core modules. Statistical analysis and result interpretation are basically reasonable.

Evaluation within the Field.

This work is positioned in the popular direction of AI-assisted medical image diagnosis, focusing on practical problems of chest CT interpretation in oncology. The proposed DVT and CL-IJLO models achieve competitive results compared with mainstream baseline methods on self-built multi-center datasets, and perform well in sensitivity which is critical for clinical screening. The research verifies the feasibility of using routine chest CT for incidental thyroid cancer screening and longitudinal nodule risk assessment. Nevertheless, the datasets are limited in total sample size, and the model generalization across more types of CT devices and patient groups is not fully validated.

Completeness and Coherence.

The thesis has a complete structural layout, covering research background, literature review, model design, dataset introduction, multi-dimensional experiments, result analysis and prospect. Each chapter follows unified writing norms, with standard formulas, charts and

indicator explanations. The overall narrative progresses logically: starting from thyroid cancer screening, then moving to longitudinal pulmonary nodule prediction, and ending with exploratory multimodal research. The correlation between each research module is clear, and professional terminology is used consistently throughout. Some experimental parameter settings and data preprocessing details are briefly described.

Conclusion.

Overall, this thesis completes the preset research objectives, with complete experimental verification and standard writing specifications. The author combines deep learning with clinical chest CT diagnosis, puts forward improved models for two cancer screening tasks, and verifies their effectiveness on multi-center data. The research has practical reference value for clinical auxiliary diagnosis. The thesis meets the requirements for a Ph.D. in Computer Applied Technology.

Additional Comments.